Graph Coloring: From Conflicts to Efficient Allocation

Chromatic number • Bounds • Greedy coloring • Welsh–Powell

One graph. Different orders. Different numbers of colors.

Central question: How many resources are sufficient — and how many are necessary?

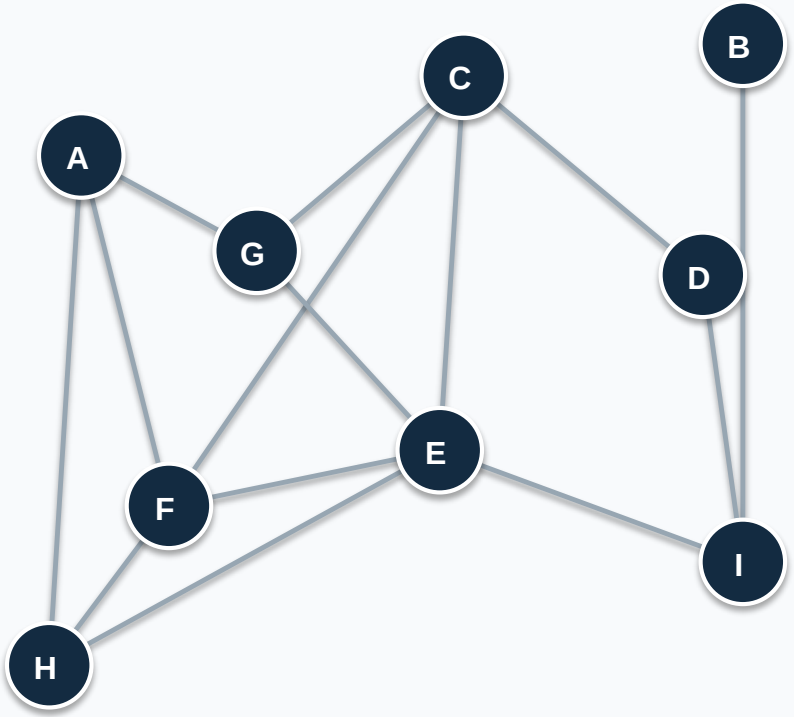
Learning outcomes and today’s route

- Translate a practical conflict into a graph-coloring model.
- Establish lower and upper bounds on the chromatic number.
- Execute first-fit greedy coloring for a specified vertex order.
- Compare two orders and distinguish feasible from optimal coloring.
- Interpret colors as time slots, channels or shared resources.

Case study: Nine examinations, shared students

A conflict edge means at least one student is enrolled in both exams.

- Vertices A–I represent nine examinations.
- Adjacent exams cannot occupy the same time slot.
- Non-adjacent exams may share a time slot.
- Goal: minimize the number of slots without clashes.



We will reuse this SAME graph to test the bounds and both algorithms.

From a valid coloring to an optimal coloring

Proper coloring

A mapping $c: V \to \{1,\dots,k\}$ such that $c(u) \neq c(v)$ for every edge $uv \in E$.

Chromatic number

$\chi(G)$ = minimum k for which G admits a proper k-coloring.

A 4-coloring proves $\chi(G) \leq 4$; it does NOT prove $\chi(G) = 4$.

Recognize chromatic numbers from structure

Complete graph K_n

$\chi(K_n) = n$

Every pair conflicts.

Cycle C_n

$\chi = 2$ if n even; 3 if n odd

Odd cycles obstruct 2-coloring.

Bipartite graph

$\chi = 2$ if it has an edge

Two independent parts.

Edgeless graph

$\chi = 1$ if V is nonempty

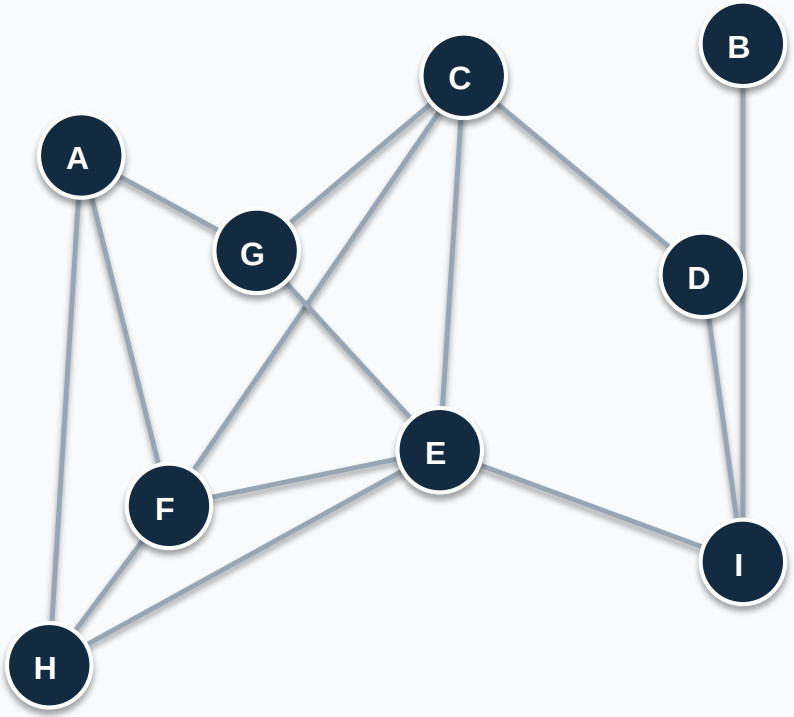
No conflicts.

Lower bound: a clique forces distinct colors

In the exam graph, C–E–G–C is a triangle.

Those three examinations must occupy three different slots.

$$\omega(G) \le \chi(G)$$



A clique certifies a minimum number of colors; it does not construct a full coloring.

Upper bound: why Δ(G) + 1 colors suffice

When a vertex is colored, it has at most Δ(G) colored neighbors.

Among Δ(G) + 1 available colors, at least one is unused by those neighbors.

$$\omega(G) \le \chi(G) \le \Delta(G) + 1$$

Exam graph: $\omega(G) \ge 3$; $\Delta(G) = 5 \rightarrow 3 \le \chi(G) \le 6$.

Greedy first-fit coloring: algorithmic idea

- Choose an ordering $v_1, v_2, \ldots, v_n$.
- Process vertices in that order.
- For each vertex, inspect the colors already assigned to its neighbors.
- Assign the smallest positive color not used by those neighbors.
- A proper coloring is guaranteed; a minimum coloring is not.

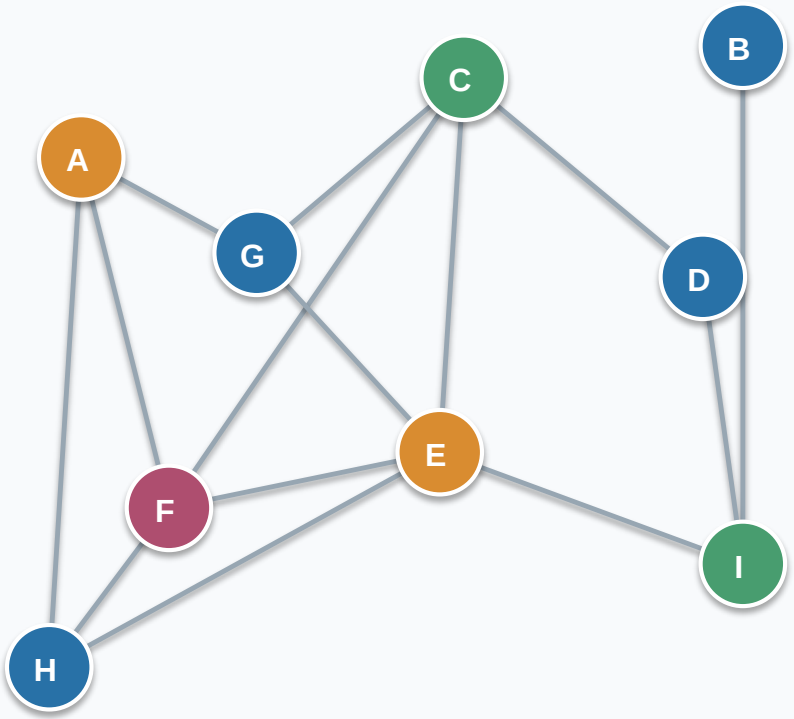
With adjacency lists and a suitable color-marking array: O(V + E) for a fixed ordering.

Practical graph: first-fit with an unfavorable order

Order: B, H, D, E, I, A, C, F, G

- Color 1: B, H, D, G
- Color 2: E, A
- Color 3: I, C
- Color 4: F

4 time slots

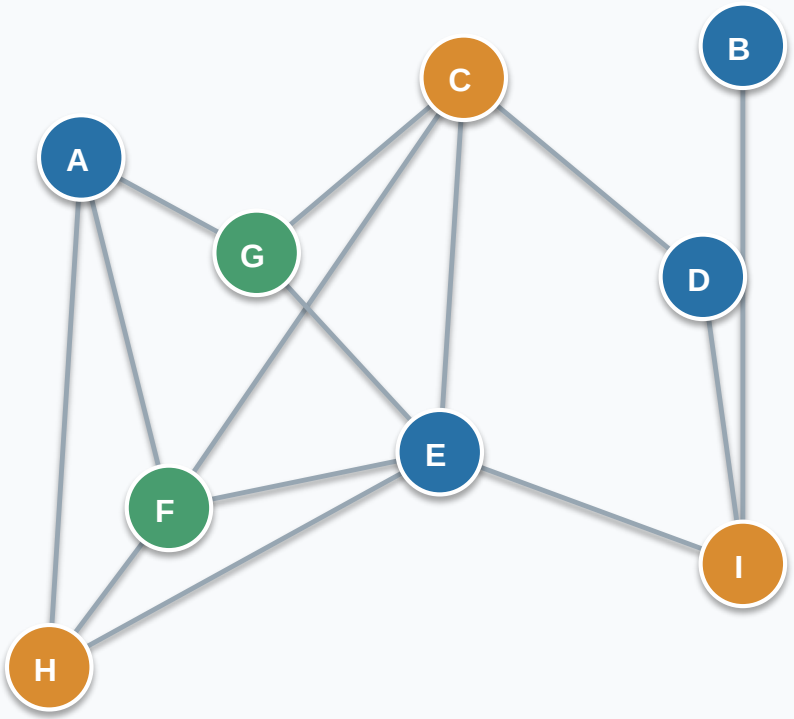


Degree-ordered greedy: Welsh–Powell-style heuristic

Order: E, C, F, A, G, H, I, D, B

- Color 1: E, A, D, B
- Color 2: C, H, I
- Color 3: F, G

3 time slots



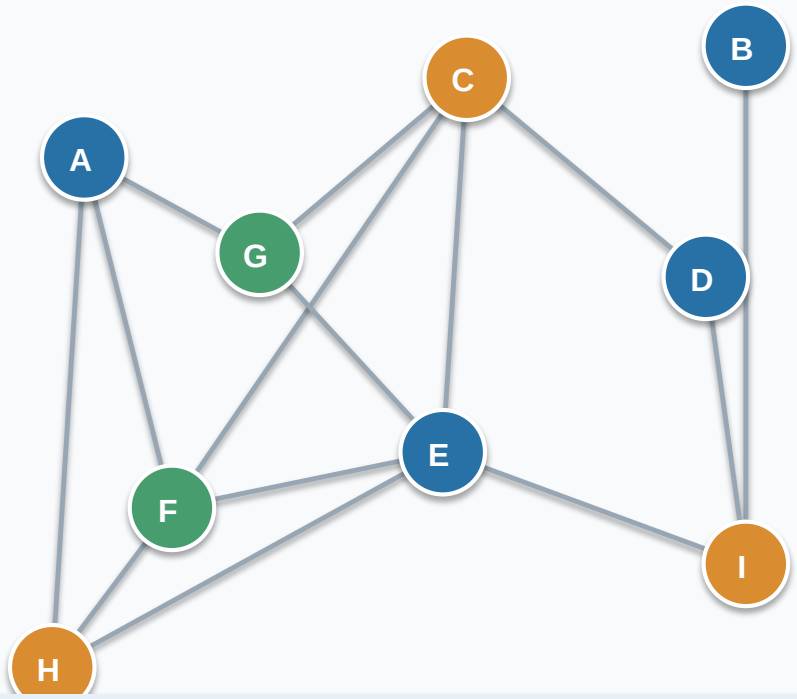
Why three slots are OPTIMAL for this graph

Lower bound

Triangle C–E–G requires at least 3 colors.

Upper bound

We have explicitly constructed a valid 3-coloring.



$$3 \le \chi(G) \le 3 \quad \Rightarrow \quad \chi(G) = 3$$

What changed? Same graph, different vertex order

Unfavorable first-fit

B, H, D, E, I, A, C, F, G

4 colors

Degree-ordered first-fit

E, C, F, A, G, H, I, D, B

3 colors

Vertex ordering is a heuristic choice — it can change the answer, not the underlying χ(G).

Welsh–Powell: implementation detail

- Sort vertices by non-increasing degree.
- In the first-fit variant, assign the smallest permissible color in that order.
- The original color-class formulation repeatedly selects a color and scans the order for mutually nonadjacent uncolored vertices.
- Both are degree-based heuristics; neither guarantees $\chi(G)$.

For our example, the displayed degree order gives a valid optimal 3-coloring.

Practical application 2: wireless frequency assignment

Interfering transmitters → adjacent vertices

- Vertex: access point or transmitter.
- Edge: nearby devices cannot reuse the same channel.
- Color: frequency channel.
- A proper coloring avoids modeled interference; $\chi(G)$ is the minimum number of channels under this conflict model.

Real deployments may require weighted interference, channel separation and signal-strength constraints.

Practical application 3: compiler register allocation

Variables live at the same time cannot share a register.

- Vertex: program variable or temporary value.
- Edge: two variables are simultaneously live.
- Color: physical register.
- If available registers are insufficient, some values may need to be spilled to memory.

Graph coloring is a model of register allocation, not the entire compiler optimization problem.

Class task: prove, don’t just color

Use the nine-exam conflict graph from today.

- Find a triangle and infer a lower bound on $\chi(G)$.
- Run first-fit on order B, H, D, E, I, A, C, F, G.
- Run degree-ordered first-fit (break equal-degree ties alphabetically).
- Give a short proof that the smaller coloring is optimal.

Challenge: Can you produce another ordering that uses exactly 3 colors?

Takeaways and next lecture

- Coloring translates conflicts into constrained resource assignment.
- Clique structure yields lower bounds; maximum degree yields an upper bound.
- Greedy first-fit is fast, but ordering affects the number of colors.
- A matching lower bound and constructed coloring certify optimality.
- Next: chromatic polynomial, edge coloring and the planarity–coloring connection.